

Rafael Figueroa Gomez

Seattle, WA | rafaelrfg05@gmail.com | 360-421-1981 | <https://www.linkedin.com/feed/>

PROFESSIONAL SUMMARY

Electrical and Computer Engineering student at the University of Washington with hands-on experience designing and building embedded systems, analog circuits, and hardware prototypes for medical and energy applications. Proficient in Python, C++, Java, MATLAB/Simulink, KiCAD/Altium, and microcontroller-based development. Bilingual in English and Spanish with a strong track record of technical execution, cross-functional collaboration, and academic excellence.

EDUCATION

University of Washington

Bachelor of Science, Electrical and Computer Engineering.

Seattle, WA

Expected June 2028

Whatcom Community College

Associate in Science Transfer Degree, Electrical and Computer Engineering. GPA: 3.95.

Bellingham, WA

June 2026

Relevant Coursework: Advanced Multivariable Calculus, Electromagnetic Physics, Differential Equations, Linear Algebra, Java with Classes and Objects, Circuits, Engineering Numerical Methods/Computations

EXPERIENCE

Coa Mexican Eatery

2022 – 2025

LaConner / Bellingham, WA

- Trained and cross-trained staff across 3 roles, reducing onboarding time by 2 weeks
- Led a team of 3 to deliver exceptional customer service, increasing sales by 5%
- Memorized full menu allergen profiles to ensure guest safety

Peer Tutor

2025 – 2026

Whatcom Community College

Bellingham, WA

- Guided 15+ peers each quarter through Calculus, Physics, Statics, and Differential Equations using clear, structured explanations
- Created a welcoming tutoring environment that reduced student anxiety around difficult coursework
- Applied subject expertise to break down unfamiliar problems for students in real time

Peer Mentor

2025 – 2026

Whatcom Community College

Bellingham, WA

- Organized and led engineering outreach events to encourage prospective students to pursue engineering
- Coordinated with team members to ensure events were inclusive and created a positive impression among new students

PROJECTS

Negative Pressure Wound Therapy Device

- Developed an operational medical device to promote blood flow and accelerate healing, integrating Python, circuit analysis, and microcontroller-based software development
- Validated device through continuous multi-hour operation and presented findings to class; selected for encouragement to present at a state conference

Heartbeat Monitor Circuit

- Built a prototype heartbeat monitor circuit incorporating analog filters, signal measurement, 3D modeling and printing, and IC design to produce an operational ASIC

Solar Panel Energy Optimization (MATLAB/Simulink)

- Simulated multiple system parameters in MATLAB/Simulink to evaluate algorithmic approaches for solar panel maximum power point tracking, researching dynamic algorithm switching based on operating conditions
- Produced an IEEE-formatted research paper documenting methodology and findings

Blackjack Game (Java)

- Implemented a fully functional Blackjack game in Java, applying object-oriented principles including classes, inheritance, and constructors

SKILLS

Technical: KiCAD/Altium, Soldering, 3D Modeling/Printing, Onshape, Raspberry Pi Pico (Microcontroller), Multimeter/Oscilloscope

Programming: Python, C++, MATLAB/Simulink, Java

Languages: English (Native), Spanish (Fluent)